第十一讲
弯曲切应力
梁的强度条件

作业：6-13,14(b),16,18,17;

加问：若保持梁长不变，移动支座B到何处，梁上许用载荷[F]最大，是多少？

第六章　弯曲应力

6.1 对称弯曲正应力

6.2 截面几何性质（一）

6.3 对称弯曲切应力

6.4 梁的强度条件

6.5 梁的合理强度设计

6.6 双对称截面梁的非对称弯曲

第六章 弯曲应力

6.3 对称弯曲切应力

1、求解思路

2、矩形截面梁对称弯曲切应力

3、弯曲切应力对弯曲正应力的影响

4、对称薄壁截面梁弯曲切应力

5、例题

1、求解思路

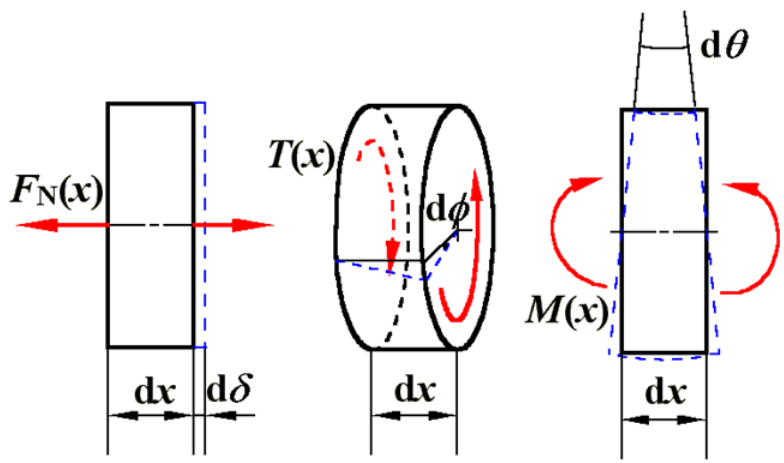
弯曲正应力求解回顾

单向受力假设

弯曲平面假设 $\longrightarrow$ $\varepsilon = \dfrac{y}{\rho}$ $\quad \sigma(y) = E\dfrac{y}{\rho}$ $\quad \int_A \sigma dA = 0$ $\quad \int_A y\sigma \mathrm{d}A = M$

求解思路：综合考虑几何、物理、静力学三方面

拉压、圆轴扭转平面假设 $\longrightarrow$ $\sigma = \dfrac{F_N}{A}$, $\quad \tau_\rho = \dfrac{T}{I_P}\rho$



三个平面假设

↓

微段变形

↙ ↘

应力

$\Delta l = \int d\delta, \Delta\varphi = \int d\varphi$

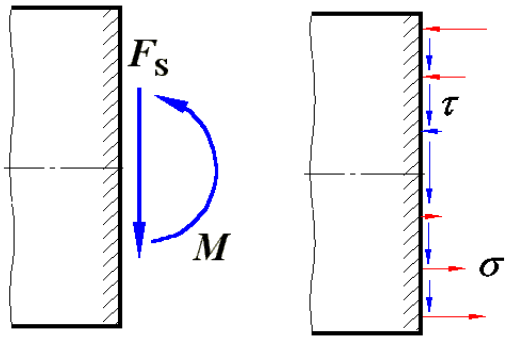
梁变形？

揭示了拉压弯扭的基本特征

$$\mathrm{d}\delta = \frac{F_N(x)}{EA}\mathrm{d}x, \quad \mathrm{d}\varphi = \frac{T(x)}{GI_P}\mathrm{d}x, \quad \mathrm{d}\theta = \frac{M(x)}{EI_z}\mathrm{d}x$$

1、求解思路

问题：已知截面剪力弯矩，求应力



弯曲切应力

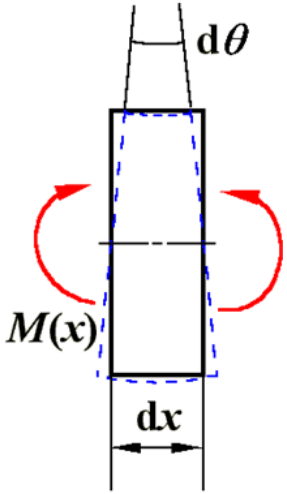
$$\int_A \tau \, \mathrm{d}A = F_S$$

可否利用弯曲平面假设？

没有切应变，何来切应力？

可否假设切应变分布？

表面变形观察看不到！



追根溯源，从实践中找切应力分析的源头。

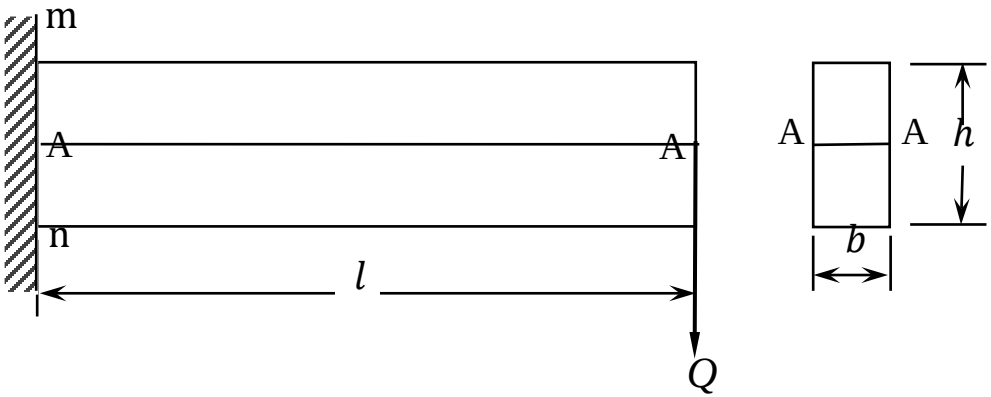
1、求解思路

儒拉夫斯基(Jourawski)的贡献

S.P. Timoshenko, Histroy of Strength of Materials, p141.

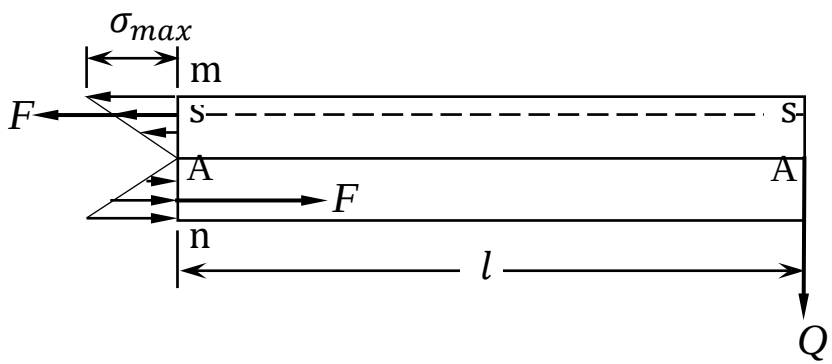
- **1844年，建造横跨维列比亚河的大桥，**
- **采用高度很大的木梁和木组合梁，**
- **木材顺纹抗剪能力很差。**

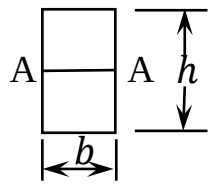
判断：梁内切应力非常重要，不能忽略。

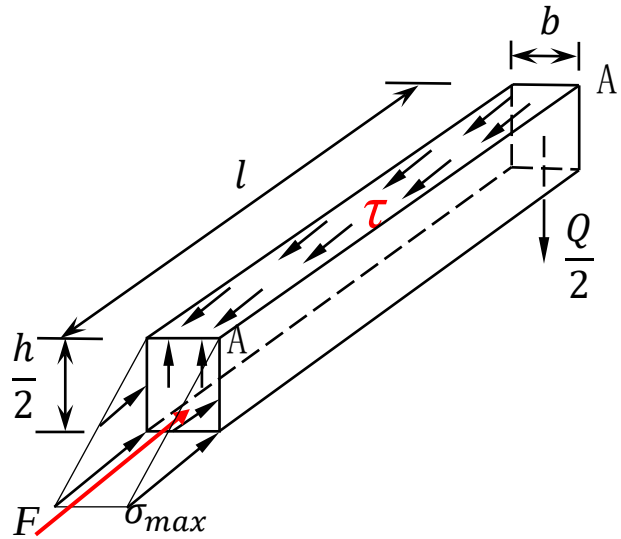


1、求解思路

儒拉夫斯基(Jourawski)的贡献







断定：固定端mn上法向应力有使AA面产生剪切的趋势。

假设：纵向截面切应力均匀分布；

启示：利用正应力的差，分析纵向截面切应力，得到横截面切应力。

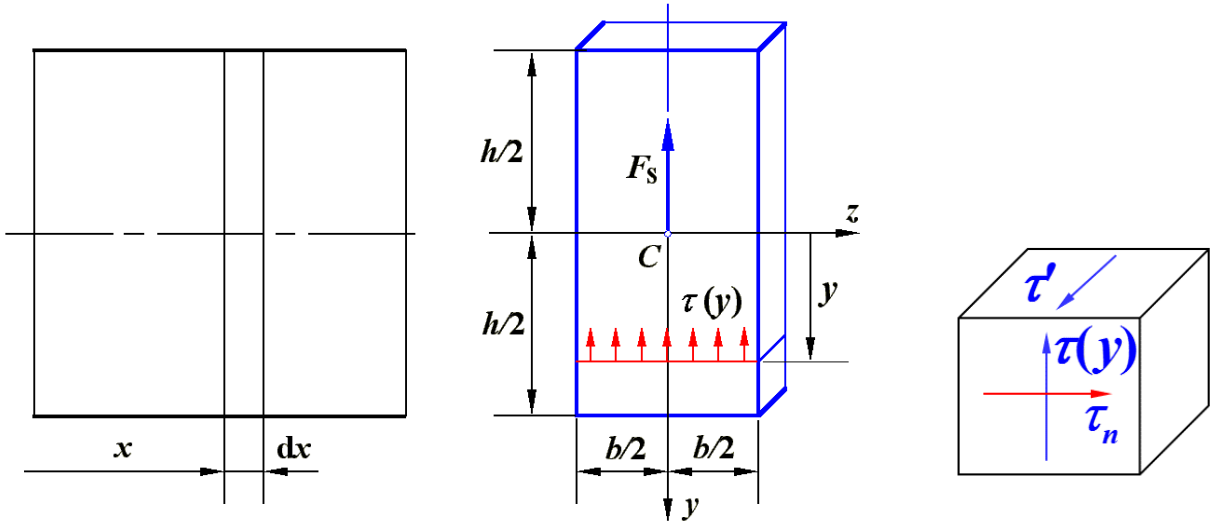
$$\sigma_{max} = \frac{6Ql}{bh^2}$$

$$F = \frac{1}{2}\sigma_{max}\frac{h}{2}b = \frac{3Ql}{2h}$$

$$\tau = \frac{F}{lb} = \frac{3}{2}\frac{Q}{bh}$$

2、矩形截面梁对称弯曲切应力

问题 狭窄矩形截面梁($h>b$)，分析其弯曲切应力分布



假设 $\tau(y)$ // 截面侧边，并沿截面宽度均匀分布

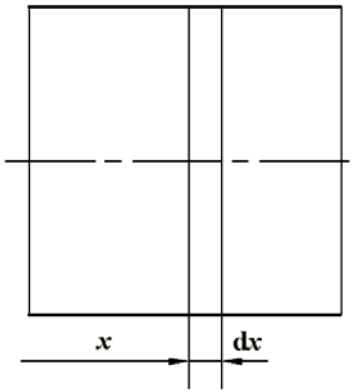
思路 由正应力的差确定切应力

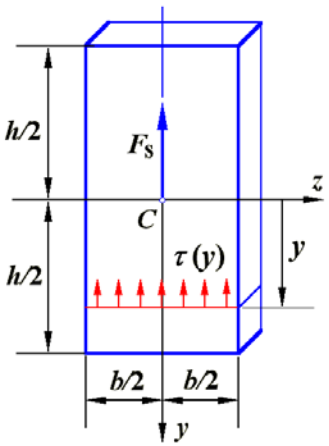
2、矩形截面梁对称弯曲切应力

取微段，再取分离体ω

分离体纵向平衡

$$\tau' b\mathrm{d}x = \mathrm{d}F$$

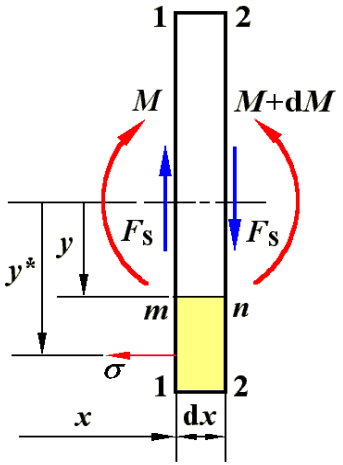


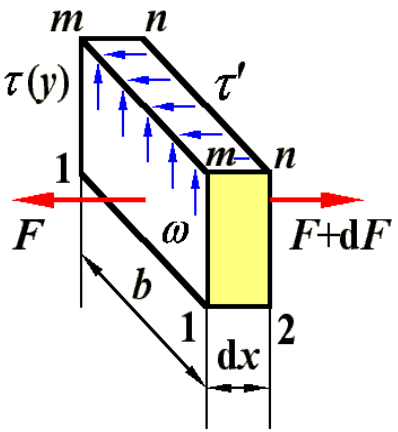


$$F = \int_\omega \sigma \mathrm{d}A = \frac{M}{I_z}\int_\omega y^* \mathrm{d}A = \frac{M\,S_z(\omega)}{I_z}$$

$$\tau(y) = \tau'(y) = \frac{S_z(\omega)}{bI_z}\frac{\mathrm{d}M}{\mathrm{d}x}$$

$$\tau(y) = \frac{F_S S_z(\omega)}{I_z b}$$





$S_z(\omega)$ —面积 ω 对中性轴 z 的静矩

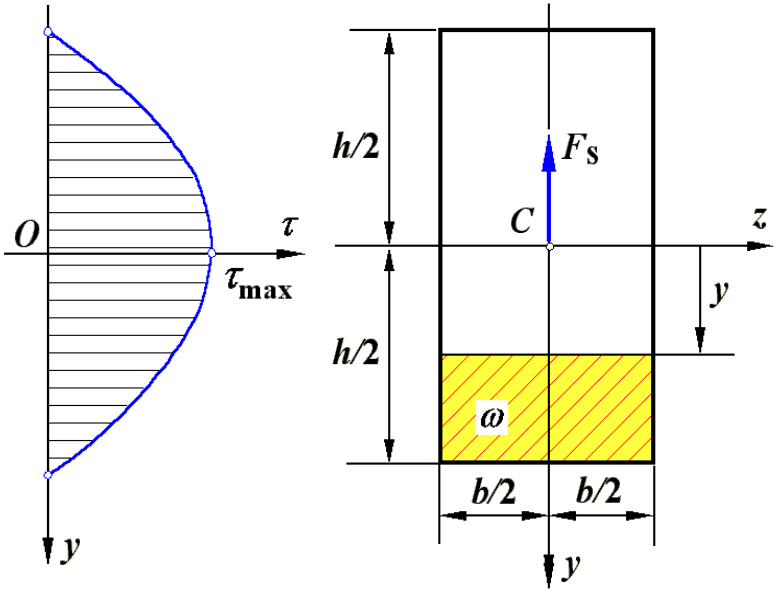
2、矩形截面梁对称弯曲切应力

切应力沿高度的变化

$$S_z(\omega)=b\left(\frac{h}{2}-y\right)\cdot\frac{1}{2}\left(\frac{h}{2}+y\right)$$

$$I_z=\frac{bh^3}{12}$$

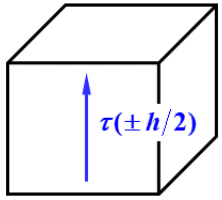
$$\boxed{\tau(y)=\frac{3F_S}{2bh}\left(1-\frac{4y^2}{h^2}\right)}$$



结果分析与验证

1) $\int_{-\frac{h}{2}}^{\frac{h}{2}}\tau(y)b\mathrm{d}y=F_S$?

2) $y=\pm\frac{h}{2},\quad \tau=0$

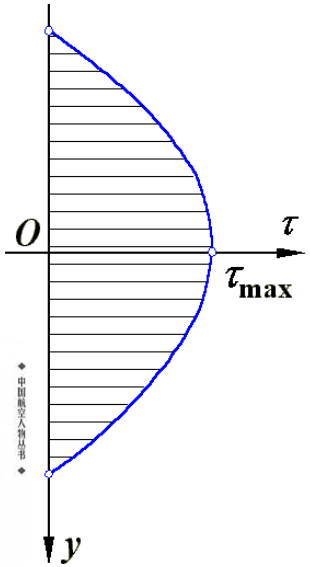


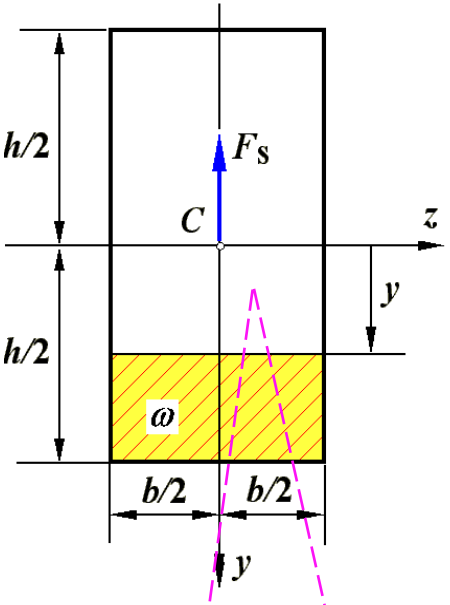
满足切应力互等定理

2、矩形截面梁对称弯曲切应力

$$\tau(y)=\frac{3F_S}{2bh}\left(1-\frac{4y^2}{h^2}\right)$$

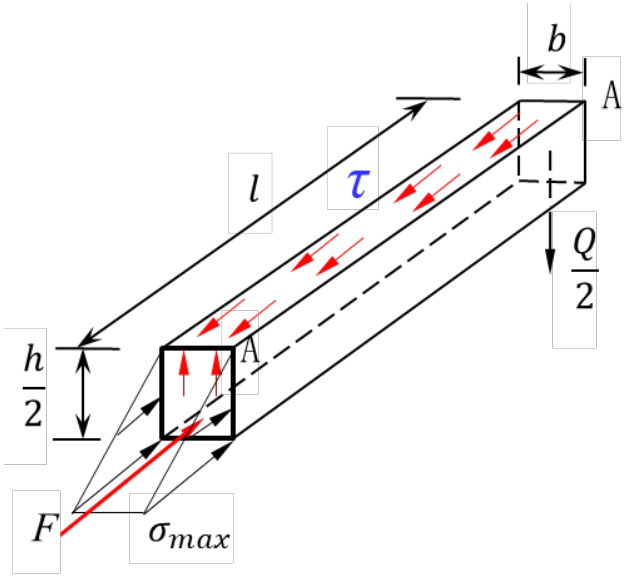
自以为非
多方求证





结果分析与验证

3）$\tau_{max}=\frac{3}{2}\frac{F_S}{A}$



工程院院士
北航教授、博导

4）试一试：
若取微段上部，
是否殊途同归？

途径：简单情况，精确理论，
殊途同归，试验验证，等等。

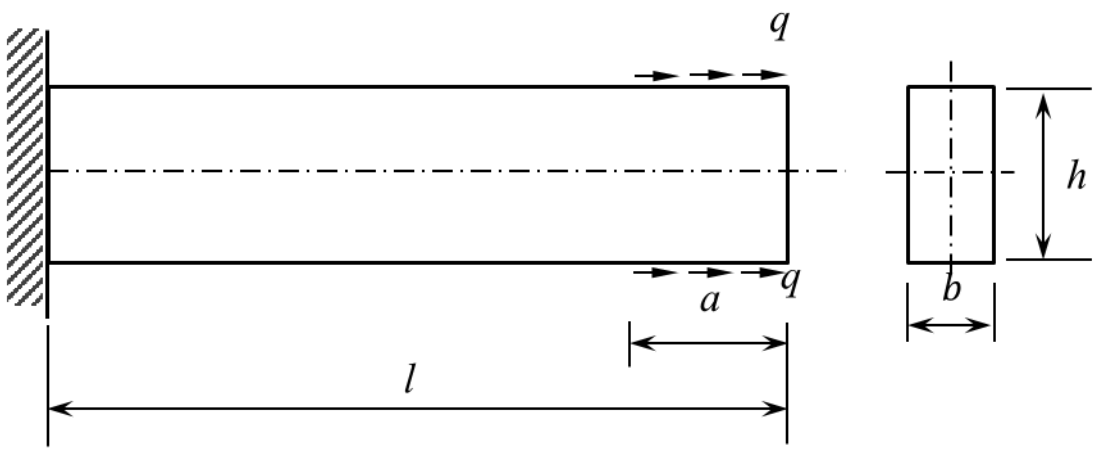
2、矩形截面梁对称弯曲切应力

思考

矩形截面杆件，受局部剪切载荷作用

1）端头区域，拉压平面假设是否成立？

2）横截面是否存在切应力？如何确定？



3、弯曲切应力对正应力的影响

切应力非均布

↓

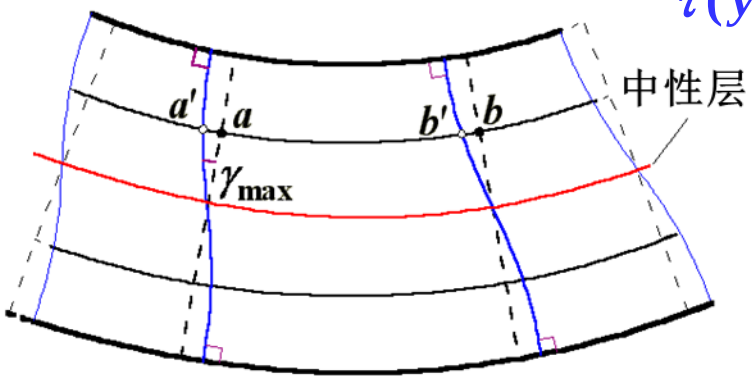
切应变非均布

↓

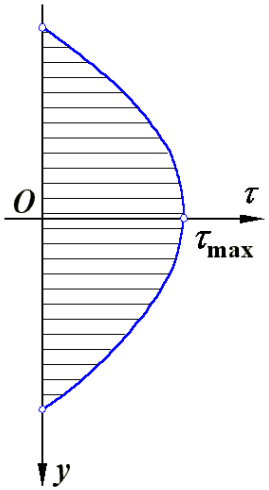
横截面翘曲

↓

平面假设不成立



$$\tau(y)=\frac{3F_S}{2bh}\left(1-\frac{4y^2}{h^2}\right)$$



$$\sigma=\frac{M}{I_Z}y$$

?

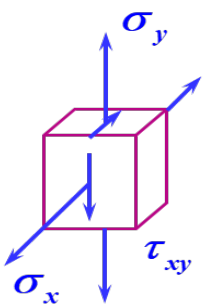
1）当梁上无分布载荷、剪力为常数时，各截面翘曲量相同，$\widehat{ab}=\widehat{a'b'}$，不影响弯曲正应力的线性分布。

2）当梁上有横向分布载荷时，各截面翘曲量不同，对正应力有影响；只要 $l>5h$（细长梁），正应力公式仍足够精确。

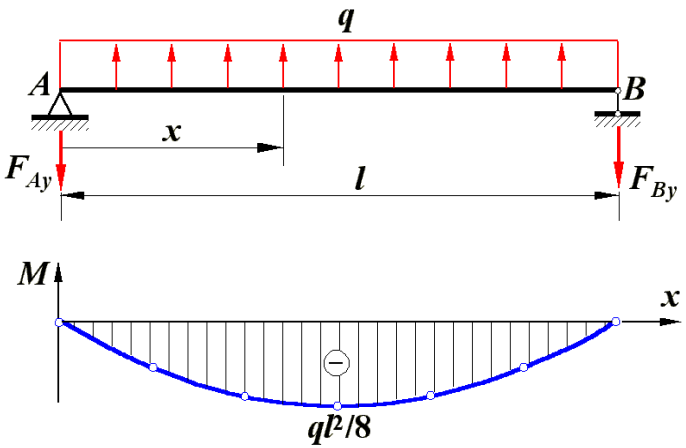
3、弯曲切应力对正应力的影响

放弃平面假设的弹性力学精确解

$$\begin{cases} \sigma_x = \dfrac{M(x)}{I_z} y + q\dfrac{y}{bh}\left(4\dfrac{y^2}{h^2} - \dfrac{3}{5}\right) \\ \sigma_y = \dfrac{q}{2b}\left(1+\dfrac{y}{h}\right)\left(1-\dfrac{2y}{h}\right)^2 \\ \tau_{xy} = -\dfrac{3F_S(x)}{2bh}\left(1-4\dfrac{y^2}{h^2}\right) \end{cases}$$



矩形截面简支梁受均布载荷



$$M(x) = 4M_{\max}\frac{x}{l}\left(1-\frac{x}{l}\right), \quad M_{\max} = \frac{ql^2}{8};$$

$$F_S(x) = -\frac{ql}{2}\left(1-\frac{2x}{l}\right)$$

1) 切应力解答与材料力学相同；

2） $\sigma_{x\max} = \dfrac{M_{\max}}{W_z} + \dfrac{q}{5b} = \dfrac{3ql^2}{4bh^2}\left(1+\dfrac{4}{15}\left(\dfrac{h}{l}\right)^2\right)$

当 $l = 5h$, $\sigma_{x\max} = \dfrac{3ql^2}{8}\dfrac{6}{bh^2}\left(1+\dfrac{16}{1500}\right) = \dfrac{M_{\max}}{W_z}$.

3） $0 \le \sigma_y \le q/b \ll \sigma_{x\max}$.

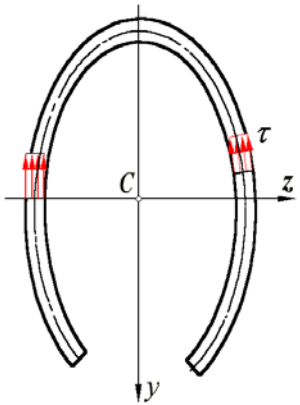
$$\frac{\sigma_{x\max}}{\tau_{xy\max}} = \frac{l}{h}\left(1+\frac{4}{15}\left(\frac{h}{l}\right)^2\right)$$

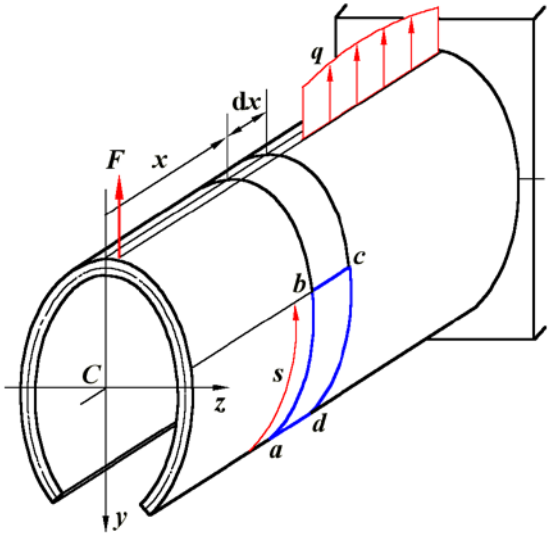
细长梁，切应力可忽略，横向无挤压

4、对称薄壁截面梁弯曲切应力

对称薄壁截面梁对称弯曲

假设 切应力平行于中心线切线沿厚度均匀分布



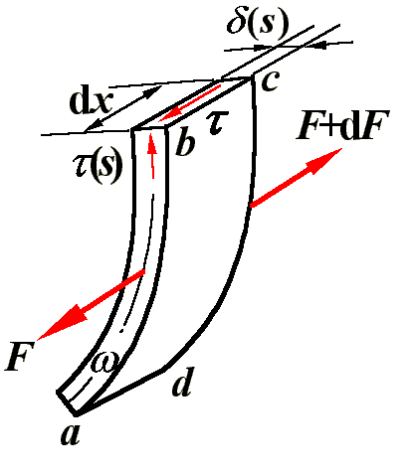


根据正应力的差确定切应力

$$\tau(s)\delta(s)\,\mathrm{d}x = \mathrm{d}F$$

$$F = \int_{\omega} \sigma \mathrm{d}A = \int_{\omega} \frac{My}{I_z}\mathrm{d}A = \frac{MS_z(\omega)}{I_z}$$

$$\tau(s) = \frac{F_S S_z(\omega)}{I_z \delta(s)}$$



S_z-截面 ω 对 z 轴的静矩；

I_z- 整个截面对 z 轴的惯性矩。

4、对称薄壁截面梁弯曲切应力

工字梁

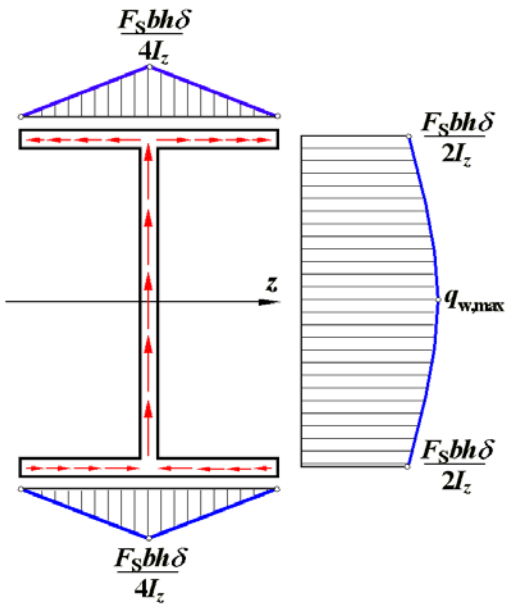
$$\tau(s)=\frac{F_S S_z(\omega)}{I_z\delta(s)}$$

翼缘剪流

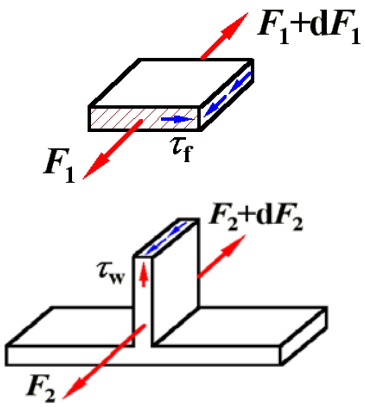
$$q_f=\frac{F_S S_z(\eta)}{I_z}=\frac{F_S}{I_z}\cdot\eta\delta\cdot\frac{h}{2}=\frac{F_S h\eta}{2I_z}$$

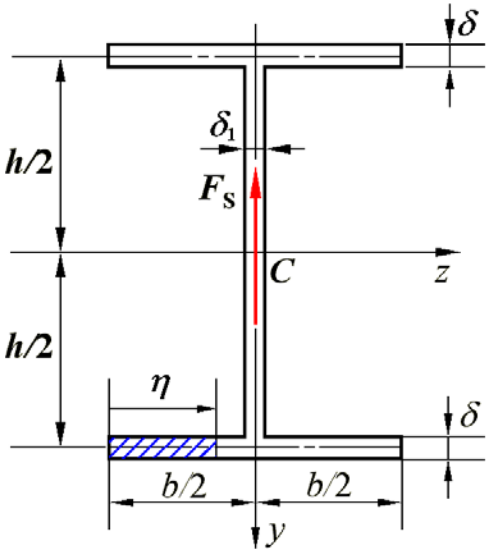
腹板剪流

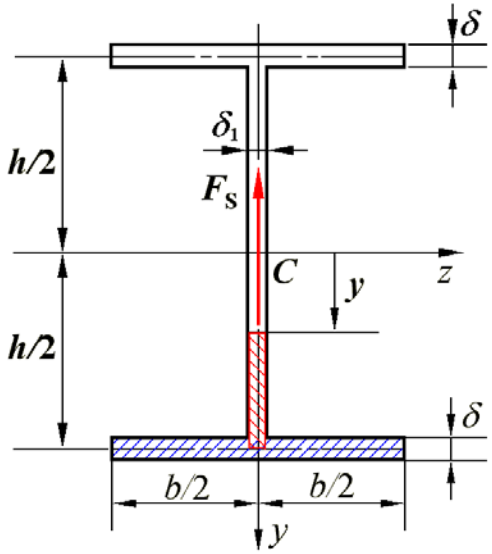
$$q_w(y)=\frac{F_S}{I_z}\left[\frac{b\delta h}{2}+\frac{\delta_1}{2}\left(\frac{h^2}{4}-y^2\right)\right]$$



$$q_{w,max}=\frac{F_S h(4b\delta+h\delta_1)}{8I_z}$$



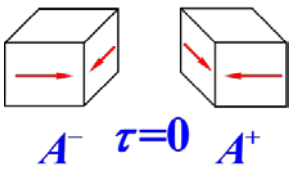




4、对称薄壁截面梁弯曲切应力

薄圆管

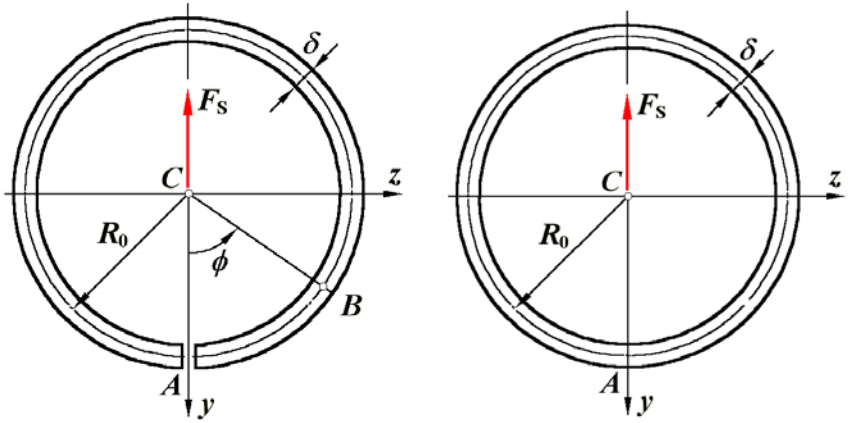
$$\tau(\varphi)=\frac{F_S S_z(\varphi)}{I_z\delta}$$



对称受力

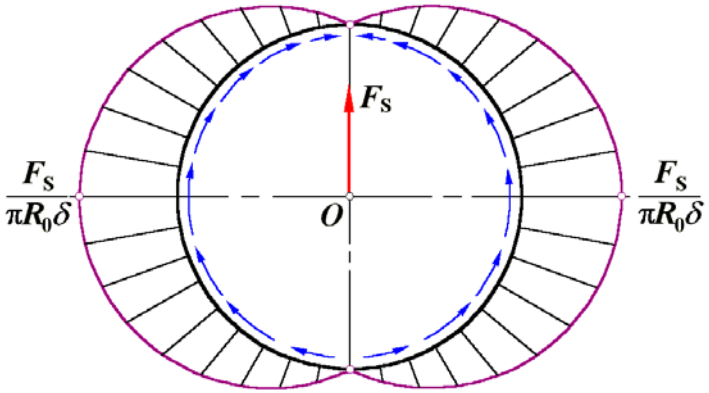
对称面

切应力为零



$$I_z=\pi R_0^3\delta$$

$$S_z(\varphi)=\int_\omega y\mathrm{d}A=\int_0^\varphi R_0\cos\alpha\cdot\delta R_0\mathrm{d}\alpha=R_0^2\delta\sin\varphi$$



$$\tau(\varphi)=\frac{F_S\sin\varphi}{\pi R_0\delta}$$

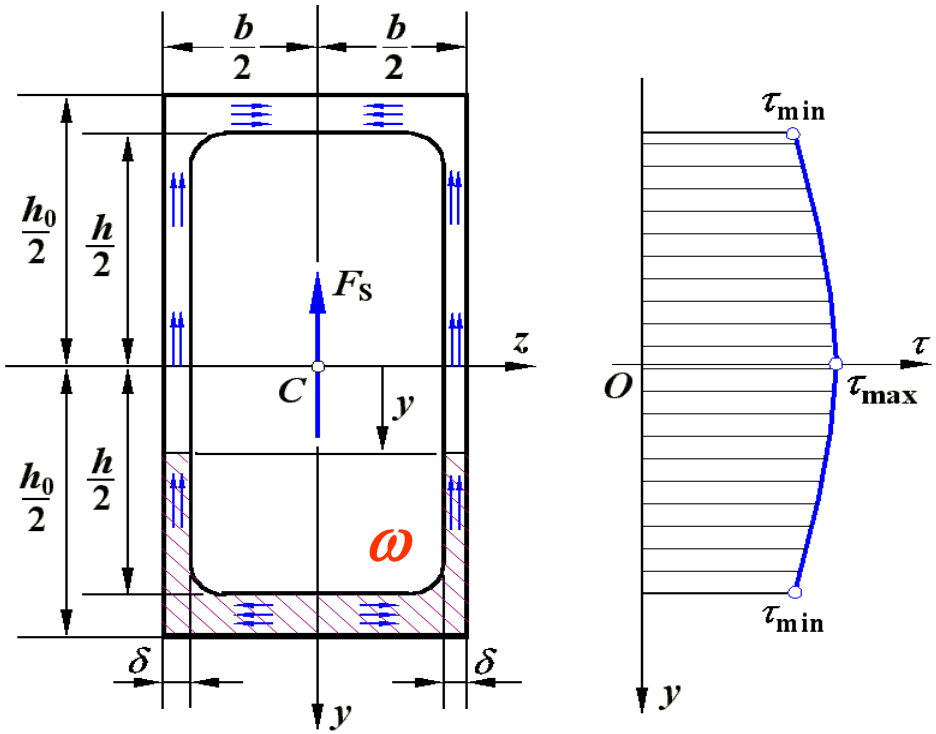
$$\boxed{\tau_{\max}=\frac{F_S}{\pi R_0\delta}}$$

4、对称薄壁截面梁弯曲切应力

盒式梁（箱型薄壁梁）

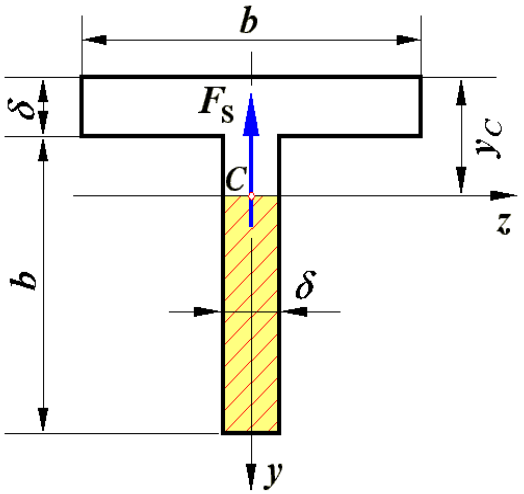
$$\tau(y)=\frac{F_S S_z(\omega)}{I_z\cdot 2\delta}$$

$$=\frac{F_S}{16I_z\delta}\left[b(h_0^2-h^2)+2\delta(h^2-4y^2)\right]$$



5、例题

例6-4 图示T形截面梁， $F_S = 15\ \text{kN}$， $I_z = 8.84\times10^{-6}\ \text{m}^4$， $b = 120\ \text{mm}$， $\delta = 20\ \text{mm}$， $y_C = 45\ \text{mm}$，求 τ_{max}、腹板与翼缘交接处切应力 τ_a



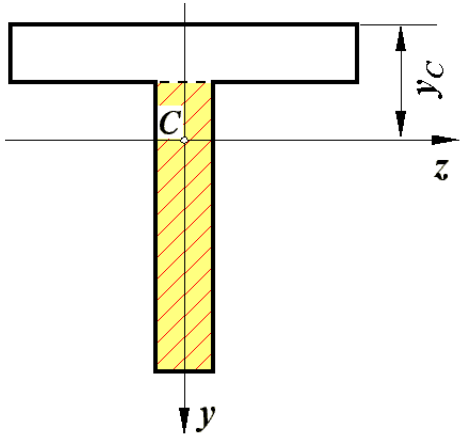
解：

$$S_{z,\max} = \frac{\delta(\delta + b - y_C)^2}{2} = 9.03\times10^{-5}\ \text{m}^3$$

$$\tau_{\max} = \frac{F_S S_{z,\max}}{I_z \delta} = 7.66\ \text{MPa}$$

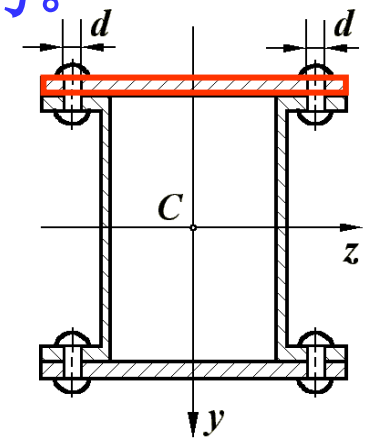
$$S_{z,a} = b\delta\left(\frac{b}{2} + \delta - y_C\right) = 8.40\times10^{-5}\ \text{m}^3$$

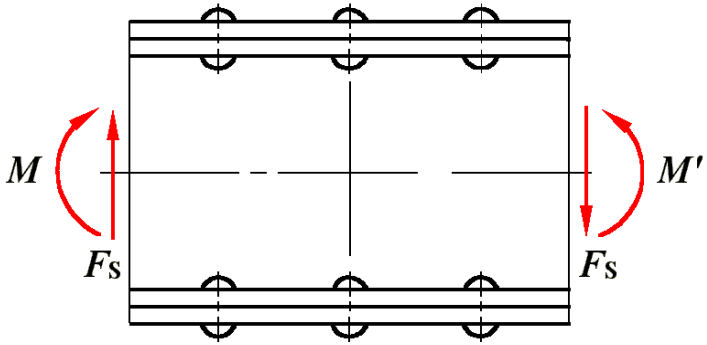
$$\tau_a = \frac{F_S S_a}{I_z \delta} = 7.13\ \text{MPa}$$



5、例题

例6-5 图示薄壁梁，上下翼板与两侧腹板铆接。已知梁段剪力 F_S，分析铆钉受力。





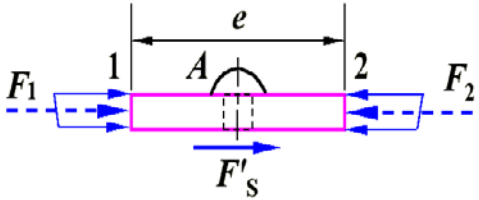
解： $2F'_S = F_2 - F_1$

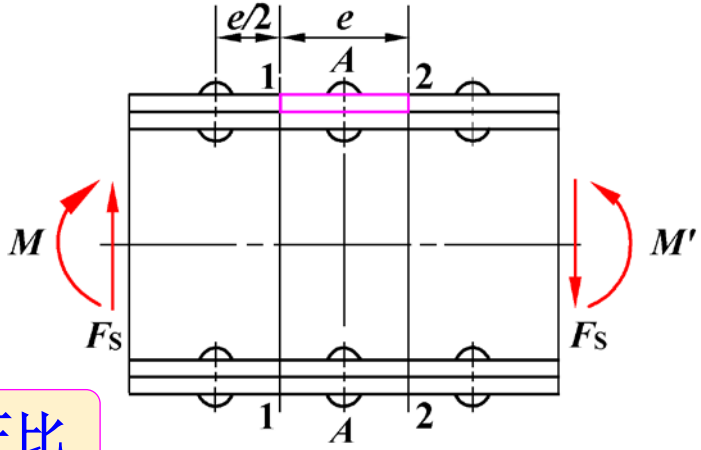
$$F_1 = \frac{M_1 \bar{S}_z}{I_z} \qquad F_2 = \frac{M_2 \bar{S}_z}{I_z}$$

$$M_2 = M_1 + F_S e$$

$$F'_S = \frac{F_S e \bar{S}_z}{2I_z}$$

$\bar{S}_z$- 上翼板对中性轴 z 的静矩





铆钉力与剪力、钉距成正比

第六章 弯曲应力

6.4 梁的强度条件

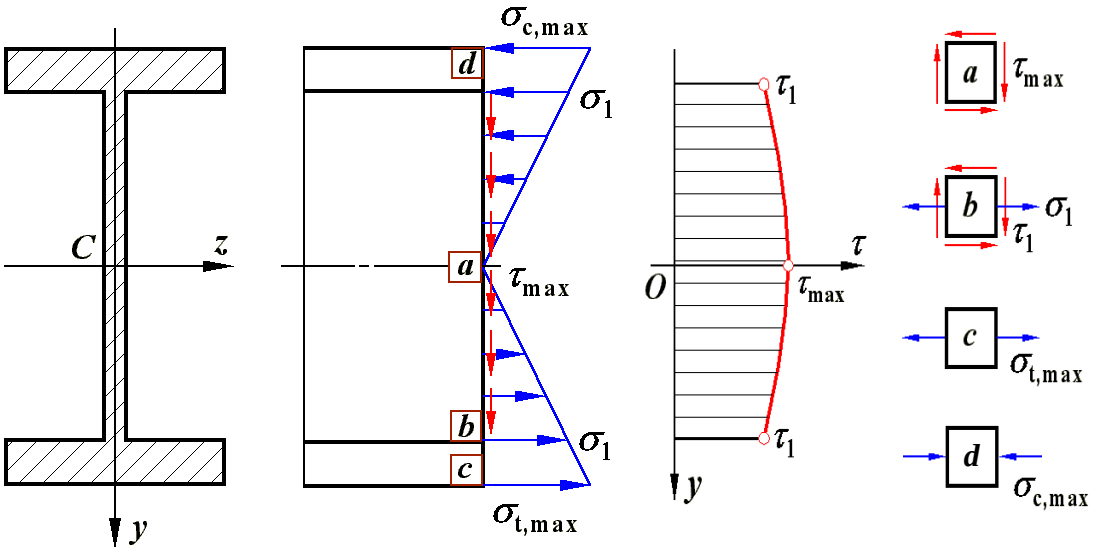
1、危险点应力状态

2、梁弯曲强度条件

3、例题

1、危险点应力状态

薄壁截面梁



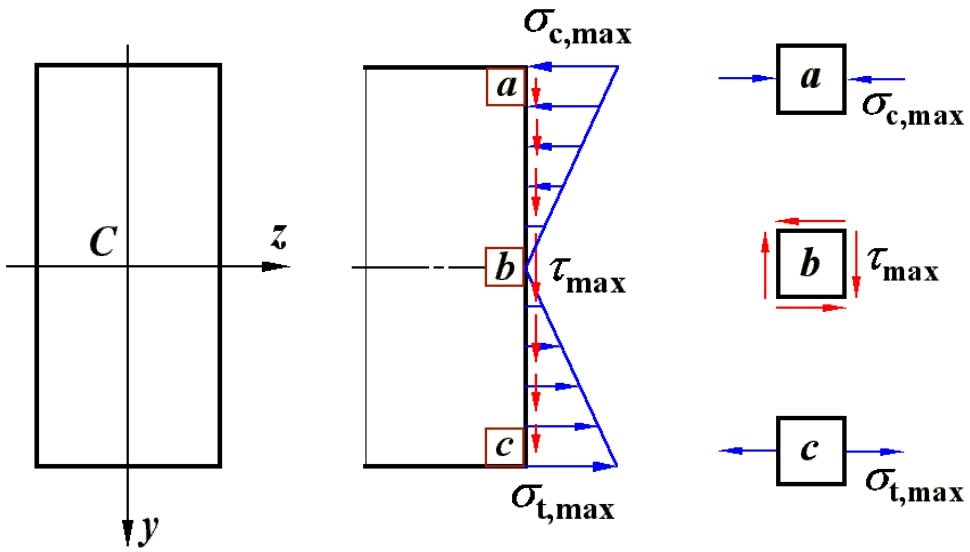
a 点处—纯剪切

b 点处—σ,τ联合作用

c , *d* 点处—单向应力

1、危险点应力状态

实心与非薄壁截面梁



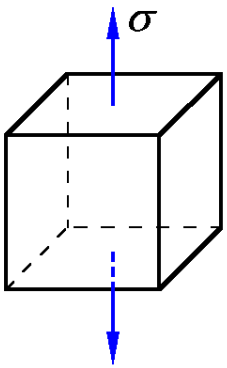
a, c 点处—单向应力

b 点处—纯剪切

2、梁弯曲强度条件

根据单向拉伸试验

弯曲正应力强度条件



$$\sigma_{\max} \le [\sigma]$$

$$\frac{M_{\max}}{W_z} \le [\sigma], \quad \left(\frac{M}{W_z}\right)_{\max} \le [\sigma]$$

$[\sigma]$ ——（塑性）材料的许用应力

$$\sigma_{t,\max} \le [\sigma_t],$$

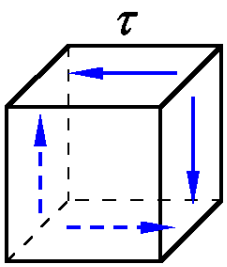
$$\sigma_{c,\max} \le [\sigma_c]$$

$[\sigma_t]$、$[\sigma_c]$

——（脆性）材料的许用拉、压应力

2、梁弯曲强度条件

根据纯剪切试验
弯曲切应力强度条件

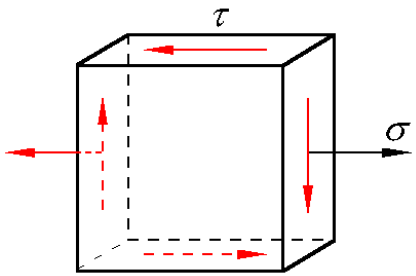


$$\tau_{\max} \le [\tau]$$

$[\tau]$ ——材料的许用切应力

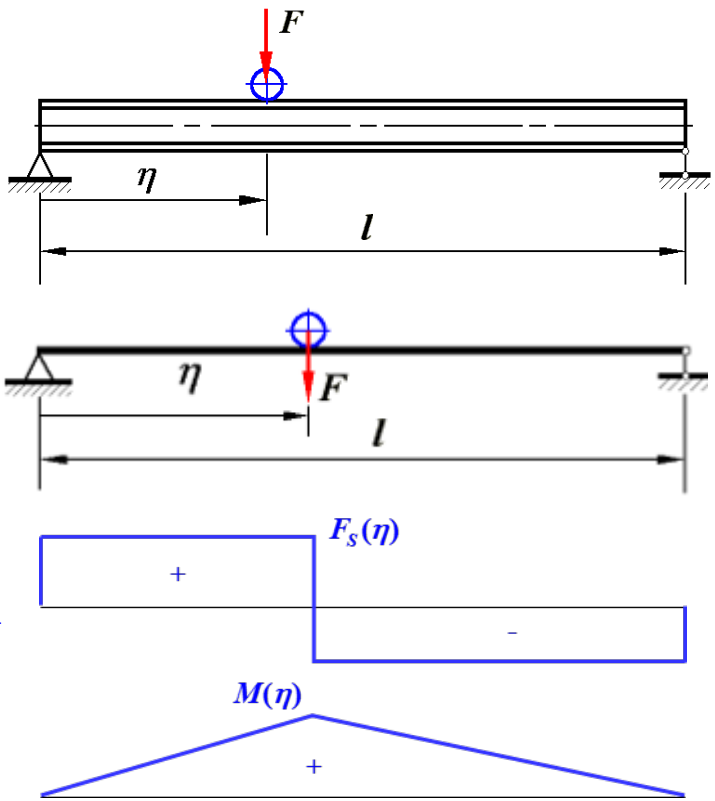
正应力和切应力联合作用处

- 不能根据试验建立强度条件；
- 需要利用应力状态分析理论与强度理论。（见第八、九章）



3、例题

例6-6 图示由工字钢制作的简易吊车，移动载荷F = 20 kN，l = 6 m，[σ] = 100 MPa，[τ] = 60 MPa，选择工字钢型号。



解：1）内力分析

$$F_{S,\max} = F_S(0) = F \qquad M_{\max} = \frac{Fl}{4}$$

2）按弯曲正应力强度条件选工字钢型号

$$W_z \geq \frac{Fl}{4[\sigma]} = \frac{20\times10^3\times6\times10^3}{4\times100} = 3\times10^5\,\text{mm}^3 = 300\ \text{cm}^3$$

查表，选 №22a

$$W_z = 309\ \text{cm}^3,\quad \delta = 7.5\ \text{mm},\quad I_z/S_{z\max} = 18.9\ \text{cm}$$

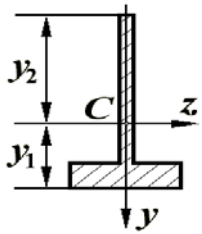
3）校核梁的剪切强度

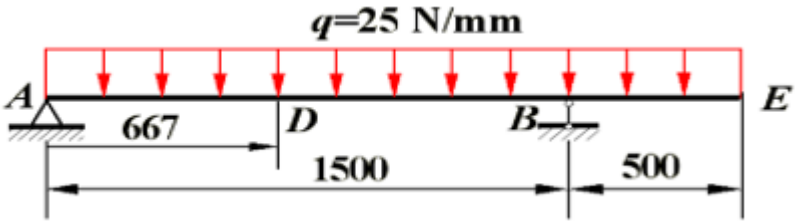
$$\tau_{\max} = \frac{F}{\dfrac{I_z}{S_{z,\max}}\delta} = \frac{20\times10^3}{189\times7.5} = 14.11\ \text{MPa} < [\tau]$$

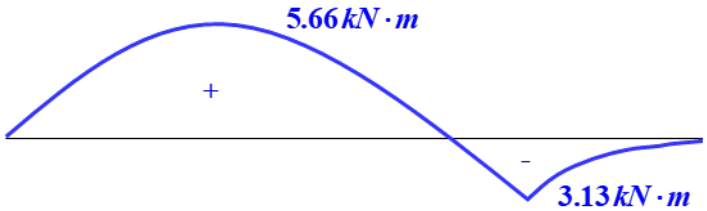
No. 22a工字钢是合适的

3、例题

例6-7 T形截面铸铁外伸梁，I_z=8.84×10^{6}mm^4，y_1=45mm，y_2=95mm，[σ_t]=35MPa，[σ_c]=40MPa。校核正应力强度。







解：危险截面：*D, B*

最大正、负弯矩

$$M_D = 5.66\ kN\cdot m,$$

$$M_B = 3.13\ kN\cdot m$$

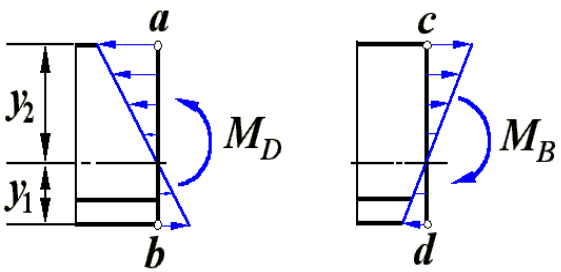
危险点 - *a, b, c*

梁的强度合格

$$\sigma_a = \frac{M_D y_2}{I_z} = 59.8\ \text{MPa} < [\sigma_c]$$

$$\sigma_b = \frac{M_D y_1}{I_z} = 28.3\ \text{MPa} < [\sigma_t]$$

$$\sigma_c = \frac{M_B y_2}{I_z} = 33.6\ \text{MPa} < [\sigma_t]$$



梁应力典型问题

铸铁梁、截面上下不对称、有正负弯矩

两个危险截面，三个危险点

本讲：对称弯曲切应力与强度条件

根据正应力差求切应力

下讲：对称弯曲切应力